Arithmetic Progressions (AP)

Chapter 5 - Hinglish Notes, Formulas, Examples + Solutions

Ye Chapter 5 (Arithmetic Progressions) ka complete guide hai. Pehle red-page ke core basics (sequence, term, common difference, a/d/n/an/Sn), fir chapter ka main content - nth term aur sum ke formulas. End me solved examples aur 20 practice questions ke step-by-step solutions hain.

EXAM * aise green tag wale topics exams me baar-baar aate hain

PART A - Core Basics (Miss Mat Karna)

1. Sequence Kya Hai?

EXAM * Exam me CONSTANTLY aata hai

Definition: Numbers ki list jo ek pattern me ho.

- Jaise 2, 4, 6, 8, ... ya 1, 4, 9, 16, ...

2. Term

EXAM * Exam me CONSTANTLY aata hai

Definition: Sequence ka har number ek 'term' hota hai.

- Pehla term, doosra term, ... nth term.

3. Difference Nikaalna

EXAM * Exam me CONSTANTLY aata hai

Rule: Do terms ka antar = baad wala - pehle wala.

- Example: 5, 9 -> 9 - 5 = 4. (Antar negative bhi ho sakta hai.)

4. Variable / Formula Samajhna

EXAM * Exam me CONSTANTLY aata hai

- a = pehla term (first term)
- d = common difference
- n = term number
- an = nth term (n-wa term)
- Sn = pehle n terms ka sum

5. Substitution

EXAM * Exam me CONSTANTLY aata hai

Matlab: Formula me values daal kar calculate karna.

- an = a + (n-1)d me a, d, n ki values daal do.

6. Basic Algebra (Do Equations)

- Kabhi a aur d ke liye 2 equations milti hain.
- Unhe subtract karke ek variable gayab (eliminate) kar do, fir solve karo.

7. Negative Numbers (Ghatti AP)

- Agar d negative ho to sequence ghatti (decrease) hoti hai.
- Example: 10, 7, 4, 1, ... (d = -3).

PART B - Chapter 5 ka Main Content

8. Arithmetic Progression (AP) Kya Hai?

EXAM * Exam me CONSTANTLY aata hai

Definition: Aisi sequence jisme har do lagatar terms ka antar (difference) HAMESHA same ho. Usi same antar ko common difference (d) kehte hain.

- Example: 3, 7, 11, 15, ... (har baar +4, to d = 4).
- General AP: a, a+d, a+2d, a+3d, ...

9. Common Difference (d)

EXAM * Exam me CONSTANTLY aata hai

Formula: d = a2 - a1 = a3 - a2 (kisi bhi term me se uska pichla term ghatao).

- Agar d same nahi aaye to wo AP nahi hai.

10. nth Term ka Formula

EXAM * Exam me CONSTANTLY aata hai

> an = a + (n - 1) d
>
> a = pehla term, d = common difference, n = term number.

- Example: 2, 5, 8, ... ka 10-wa term = 2 + (10-1) x 3 = 29.

11. Sum of First n Terms

EXAM * Exam me CONSTANTLY aata hai

> Sn = n/2 [2a + (n - 1) d]
>
> Agar last term 'l' pata ho to:
> Sn = n/2 (a + l)

- Example: 1 + 2 + 3 + ... + 100 = 100/2 (1 + 100) = 5050.

12. AP Hai ya Nahi - Kaise Check Karein

- Lagatar terms ka difference nikalo: a2-a1, a3-a2, a4-a3 ...
- Agar saare difference same hain -> AP hai; warna nahi.

13. Word Problems

- Pehle a (first term) aur d (common difference) pehchaano.
- Fir question ke hisaab se an ya Sn ka formula lagao.
- Common: savings/salary badhna, seats in rows, etc.

Quick Revision

> AP = constant common difference (d).
> d = baad wala term - pehle wala term.
> nth term: an = a + (n-1)d.
> Sum: Sn = n/2 [2a + (n-1)d] ya n/2 (a + l).
> d positive -> badhti AP; d negative -> ghatti AP.

Solved Examples - Samjho Kaise Solve Karte Hain

Neeche 10 examples step-by-step solve karke dikhaye gaye hain.

Example 1: AP 3, 7, 11, 15 ... ka common difference?

- Step: d = baad wala - pehle wala.
- Step: d = 7 - 3 = 4.

Answer:d = 4

Example 2: 2, 5, 8, ... ka 10-wa term (a10)?

- Step: a = 2, d = 5 - 2 = 3, n = 10.
- Step: an = a + (n-1)d = 2 + (10-1) x 3.
- Step: = 2 + 27.

Answer:a10 = 29

Example 3: 5, 8, 11, ... ka kaunsa term 50 hai?

- Step: a = 5, d = 3. an = 50 rakho.
- Step: 5 + (n-1)3 = 50 -> (n-1)3 = 45.
- Step: n - 1 = 15 -> n = 16.

Answer:16-wa term

Example 4: 2, 4, 6, ... ke pehle 10 terms ka sum?

- Step: a = 2, d = 2, n = 10.
- Step: Sn = n/2[2a + (n-1)d] = 10/2[2x2 + 9x2].
- Step: = 5[4 + 18] = 5 x 22.

Answer:S10 = 110

Example 5: 1 + 2 + 3 + ... + 100 ka sum?

- Step: a = 1, last term l = 100, n = 100.
- Step: Sn = n/2 (a + l) = 100/2 (1 + 100).
- Step: = 50 x 101.

Answer:5050

Example 6: Kya 1, 4, 9, 16 ek AP hai?

- Step: Differences: 4-1=3, 9-4=5, 16-9=7.
- Step: Difference same nahi (3, 5, 7).

Answer: Nahi (AP nahi hai)

Example 7: 10, 7, 4, 1 ka common difference?

- Step: d = 7 - 10 = -3 (negative).

Answer:d = -3 (ghatti AP)

Example 8: Ek AP ka 3rd term 5 aur 7th term 9 hai. a aur d?

- Step: a + 2d = 5 ... (i)
- Step: a + 6d = 9 ... (ii)
- Step: (ii)-(i): 4d = 4 -> d = 1.
- Step: a + 2(1) = 5 -> a = 3.

Answer:a = 3, d = 1

Example 9: a = 7, d = -2 -> a5?

- Step: an = a + (n-1)d = 7 + (5-1)(-2).
- Step: = 7 - 8.

Answer:a5 = -1

Example 10 (Word): Pehle mahine Rs 100 bachat, har mahine Rs 50 zyada. 12-we mahine?

- Step: a = 100, d = 50, n = 12.
- Step: an = 100 + (12-1) x 50 = 100 + 550.

Answer: Rs 650

Practice Questions

Pehle khud solve karne ki koshish karo. Neeche har question ka step-by-step solution diya gaya hai.

Q1. AP (Arithmetic Progression) kya hota hai?

Q2. AP me common difference (d) kaise nikalte hain?

Q3. nth term ka formula kya hai?

Q4. Sum of n terms ka formula kya hai?

Q5. 2, 5, 8, 11 ka common difference kya hai?

Q6. 3, 7, 11, ... ka 10-wa term kya hai?

Q7. 1, 2, 3, ..., 50 ka sum kya hai?

Q8. Kya 2, 4, 8, 16 ek AP hai?

Q9. 10, 8, 6, 4 ka common difference kya hai?

Q10. a = 5, d = 3, n = 4 -> a4 kya hai?

Q11. AP 7, 10, 13, ... ka 20-wa term?

Q12. Pehle 5 even numbers (2,4,6,8,10) ka sum?

Q13. a = 3, d = 2 ho to pehle 3 terms kya honge?

Q14. d negative ho to sequence badhti hai ya ghatti?

Q15. Sum formula me 'l' (last term) ka kya matlab hai?

Q16. Last term l ke saath sum ka formula kya hai?

Q17. AP 2, 4, 6, ..., 20 ka sum kya hai?

Q18. 3, 6, 9, ... ka kaunsa term 30 hai?

Q19. a = 1, d = 1 ho to S10 kya hai?

Q20. 5, 2, -1, -4 ka common difference?

Step-by-Step Solutions

Q1. AP kya hota hai?

- Step: Aisi sequence jisme har do terms ka antar same ho.

Answer: Constant common difference wali sequence

Q2. Common difference kaise?

- Step: Kisi term me se uska pichla term ghatao.

Answer:d = a2 - a1

Q3. nth term ka formula?

- Step: Standard AP formula.

Answer:an = a + (n-1)d

Q4. Sum of n terms?

- Step: Standard sum formula.

Answer: Sn = n/2 [2a + (n-1)d]

Q5. 2,5,8,11 ka d?

- Step: 5 - 2 = 3.

Answer:d = 3

Q6. 3,7,11,... ka 10-wa term?

- Step: a=3, d=4. a10 = 3 + 9x4 = 3 + 36.

Answer:39

Q7. 1+2+...+50 ka sum?

- Step: n=50, a=1, l=50. S = 50/2(1+50) = 25x51.

Answer:1275

Q8. Kya 2,4,8,16 AP hai?

- Step: Diff: 2, 4, 8 (same nahi).

Answer: Nahi

Q9. 10,8,6,4 ka d?

- Step: 8 - 10 = -2.

Answer:d = -2

Q10. a=5,d=3,n=4 -> a4?

- Step: 5 + (4-1)3 = 5 + 9.

Answer:14

Q11. 7,10,13,... ka 20-wa term?

- Step: a=7,d=3. a20 = 7 + 19x3 = 7 + 57.

Answer:64

Q12. Pehle 5 even numbers ka sum?

- Step: a=2,d=2,n=5. S = 5/2[4 + 4x2] = 5/2[12].

Answer:30

Q13. a=3,d=2 ke pehle 3 terms?

- Step: 3, 3+2, 5+2.

Answer:3, 5, 7

Q14. d negative ho to?

- Step: Terms chhote hote jaate hain.

Answer: Ghatti (decreasing)

Q15. 'l' ka matlab?

- Step: Sequence ka aakhri term.

Answer: Last term

Q16. Last term ke saath sum?

- Step: Chhota formula.

Answer: Sn = n/2 (a + l)

Q17. 2,4,6,...,20 ka sum?

- Step: a=2,l=20,n=10. S = 10/2(2+20) = 5x22.

Answer:110

Q18. 3,6,9,... ka kaunsa term 30?

- Step: a=3,d=3. 3+(n-1)3=30 -> (n-1)=9 -> n=10.

Answer:10-wa term

Q19. a=1,d=1 -> S10?

- Step: S = 10/2[2 + 9] = 5 x 11.

Answer:55

Q20. 5,2,-1,-4 ka d?

- Step: 2 - 5 = -3.

Answer:d = -3